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PREFACE

Two threads need to be followed in tracing the development of the concept of soil sensitivity. One leads from the Swedish Geotechnical Commission, the other links us to Karl Terzaghi. The sensitivity concept is rooted close to the source of all soil mechanics but our understanding of the causes and nature of sensitivity is still far from complete. The papers in this special issue of Engineering Geology all relate to the problem of extreme sensitivity and the problems and hazards which this can cause. The problem of sensitivity is a scientific problem of great fascination but it is also a practical problem whose solution could save many lives and avoid much loss of property.

The problem of earthflows in sensitive clays has recently been reviewed, in a geomorphological context, by M.A. Carson (1977: Slope and slope processes. Progress in Physical Geography, 1: 134—142). He suggested that too much attention had perhaps been paid to the materials of extreme sensitivity, the quick clays, in particular the study of quickness at the expense of investigations into the mechanisms involved in retrogression. This is a valid point and one which deserves to be noted. From a strictly engineering point of view, however, there is no doubt that investigations into sensitivity almost invariably involve materials of extremely high sensitivities simply because these present the most acute engineering problems. Progress is being made; mapping of dangerous soils in critical areas is a very promising development from the practical point of view, and at the other end of the scale an increasing number of mineralogical determinations are indicating the importance of cementing materials in determining the properties of soils.

The papers in this special issue range from the theoretical to the practical, from mineralogy to landslide distribution but they have the common theme of sensitivity and it is hoped that they can present some further insight into this persistent problem. This journal has been a consistent supporter of quick-clay research and the series of "Recent Quick-Clay Studies" has included some very significant contributions. I hope that this special issue can maintain that excellent tradition.

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